

Cyclistic Bike-Share Analysis: Annual Members vs Casual Riders

Executive Summary

Cyclistic, a Chicago-based bike-share company, aims to increase annual memberships by converting existing casual riders into long-term subscribers. To support this business objective, a comprehensive analysis of **12 months of bike-share trip data** was conducted using **Google BigQuery** for data preparation and analysis and **Tableau** for visualization.

The analysis uncovered significant behavioral differences between annual members and casual riders. Most notably:

- **Annual members generated 3.01 million rides**, accounting for the majority of total trips.
- **Casual riders averaged 31.94 minutes per ride**, more than double the **13.37 minutes** recorded by annual members.
- **Members primarily rode during weekdays**, indicating commuter-oriented usage.
- **Casual riders were most active on weekends**, suggesting recreational and leisure-focused behavior.
- **Ridership peaked during summer months**, highlighting the influence of seasonality on customer behavior.

These findings suggest that casual riders represent a strong membership conversion opportunity. Based on the analysis, three strategic recommendations are proposed to support Cyclistic's membership growth objectives.

1. Introduction

Cyclistic operates one of Chicago's largest bike-share programs, offering customers flexible transportation options through single rides, day passes, and annual memberships. While casual riders contribute significantly to overall ridership, annual members generate greater long-term value and profitability for the company.

The primary goal of this project was to analyze rider behavior and identify meaningful differences between annual members and casual riders. Understanding these differences can

help Cyclistic develop targeted marketing campaigns that encourage membership adoption and improve customer retention.

2. Business Task

Business Question:

How do annual members and casual riders use Cyclistic bikes differently?

The insights generated from this analysis will support data-driven marketing strategies designed to convert casual riders into annual members.

3. Stakeholders

Lily Moreno – Director of Marketing

Responsible for developing and implementing membership growth strategies.

Cyclistic Marketing Analytics Team

Uses analytical insights to support marketing decisions and campaign planning.

Cyclistic Executive Leadership Team

Responsible for approving strategic initiatives and allocating organizational resources.

4. Data Source

The analysis utilized **12 months of Cyclistic trip data** obtained through Divvy's publicly available bike-share dataset.

The dataset contained information related to:

- Ride identifiers

- Ride start and end timestamps
- Bike type
- Station information
- Rider classification (Member or Casual)

The dataset was considered suitable for analysis because it was:

- ✓ Reliable
 - ✓ Original
 - ✓ Comprehensive
 - ✓ Current
 - ✓ Free of personally identifiable information
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5. Data Cleaning Process

Data preparation and cleaning were performed using **Google BigQuery**.

The following steps were completed:

Data Consolidation

- Combined all 12 monthly datasets into a single master table.

Data Quality Assessment

- Checked all variables for missing values.
- Verified that critical fields including **ride_id**, **started_at**, **ended_at**, **rideable_type**, and **member_casual** contained no missing values.

Duplicate Detection

- Performed duplicate analysis using **ride_id**.
- Confirmed that **no duplicate records existed** within the dataset.

Feature Engineering

Created additional variables including:

- ride_length
- day_of_week
- month

These variables were used to analyze duration, weekly trends, and seasonal behavior.

Data Validation

- Removed invalid rides with zero or negative durations.
- Verified consistency across all records before beginning analysis.

After cleaning and validation, the final dataset contained **5,509,830 valid bike trips**.

6. Analysis

6.1 Total Rides by Rider Type

The analysis revealed a substantial difference in ride frequency between rider groups.

Rider Type	Total Rides
Annual Members	3,014,938
Casual Riders	2,494,892

Insight

Annual members generated approximately **520,000 more rides** than casual riders.

This indicates that members rely on Cyclistic more frequently and are likely integrating bike-share services into their daily routines.

6.2 Average Ride Length

Average ride duration revealed one of the most significant behavioral differences.

Rider Type	Average Ride Duration
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Casual Riders **31.94 minutes**

Annual Members **13.37 minutes**

Insight

Casual riders spend **more than twice as long per ride** compared to annual members.

This suggests that casual riders primarily use Cyclistic for recreational activities, sightseeing, and leisure trips, while annual members use the service for practical transportation purposes.

6.3 Rides by Day of Week

Distinct weekly patterns emerged between rider groups.

Annual Members

Highest riding activity occurred during weekdays:

- Wednesday: **469,601 rides**
- Tuesday: **458,153 rides**
- Thursday: **457,588 rides**

Casual Riders

Highest riding activity occurred during weekends:

- Saturday: **550,484 rides**
- Sunday: **474,534 rides**

Insight

The data strongly suggests that:

- **Members exhibit commuter behavior**
 - **Casual riders exhibit leisure-oriented behavior**
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6.4 Monthly Usage Trends

Ridership fluctuated significantly throughout the year.

Peak Ridership Periods

Casual Riders

- July: **435,927 rides**

Annual Members

- September: **385,959 rides**
- August: **385,426 rides**

Lowest Ridership Periods

- January
- February

Insight

Bike-share usage is highly seasonal, with demand increasing during warmer months and declining during winter.

This pattern creates strategic opportunities for seasonal marketing campaigns.

6.5 Bike Type Preferences

Bike type usage differed slightly between rider groups.

Casual Riders

Bike Type	Total Rides
Classic Bike	1,253,395
Electric Bike	931,107
Docked Bike	310,390

Annual Members

Bike Type	Total Rides
Classic Bike	1,955,673

Electric Bike 1,059,264

Insight

Classic bikes were the most popular option among both rider groups, accounting for the majority of trips.

However, electric bikes represented a substantial portion of rides, suggesting increasing demand for alternative transportation options.

7. Key Findings

Key Finding #1

Annual members generated 54.7% of all rides, demonstrating higher engagement and usage frequency.

Key Finding #2

Casual riders averaged 31.94 minutes per trip, compared with **13.37 minutes for members**, indicating fundamentally different use cases.

Key Finding #3

Members displayed strong weekday riding behavior, supporting the conclusion that they primarily use Cyclistic for commuting.

Key Finding #4

Casual riders concentrated their usage during weekends, suggesting leisure and recreational riding patterns.

Key Finding #5

Ridership for both groups increased substantially during summer months, highlighting the impact of seasonality on demand.

Key Finding #6

Classic bikes remained the dominant bike type across both customer segments.

8. Recommendations

Recommendation 1: Launch Weekend Membership Campaigns

Evidence

Casual riders recorded their highest usage on Saturdays and Sundays.

Action

Develop targeted weekend membership promotions highlighting the convenience and value of annual memberships.

Expected Benefit

Higher conversion rates among highly engaged leisure riders.

Recommendation 2: Emphasize Cost Savings in Marketing Communications

Evidence

Casual riders average nearly **32 minutes per ride**, indicating frequent and prolonged usage.

Action

Create marketing campaigns demonstrating the financial benefits of annual memberships compared with repeated day-pass purchases.

Expected Benefit

Improved membership adoption among frequent casual riders.

Recommendation 3: Concentrate Marketing Efforts During Peak Seasons

Evidence

Ridership peaks during the summer months for both rider groups.

Action

Increase digital advertising, email campaigns, and app-based promotions between June and September.

Expected Benefit

Greater campaign visibility and improved membership acquisition.

9. Conclusion

This analysis successfully identified meaningful behavioral differences between Cyclistic's annual members and casual riders using **5.5 million bike-share trips** collected over a 12-month period.

The findings demonstrate that annual members primarily use Cyclistic as a practical transportation solution, while casual riders use the service for longer, leisure-oriented trips. The clear differences in ride duration, weekday usage, and seasonal riding patterns provide valuable opportunities for targeted marketing initiatives.

By focusing on weekend riders, emphasizing membership value, and increasing marketing activity during peak riding seasons, Cyclistic can strengthen its membership conversion strategy and support long-term business growth.

Tools Used

Google BigQuery

- Data Cleaning
- SQL Analysis
- Feature Engineering

Tableau

- Dashboard Design
- Data Visualization
- Insight Communication

SQL

- Data Transformation
- Aggregation Queries
- Trend Analysis